

RESEARCH ARTICLE

CarbOnto: Data Integration Toward Net Zero

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ABSTRACT Global warming and climate change have been subjects of great interest in recent years. They are understood to be related to greenhouse gas (GHG) emissions. Although agriculture suffers the consequences of these changes, it is one of the top global emitters of GHG. While it is complex in environmental, social, and economic aspects, there is a need to advance solutions for more sustainable agriculture. In the farm environment, an important step is the generation of GHG inventories. A GHG inventory is a systematic process for measuring and recording gas emissions and sequestration. However, generating inventories on rural properties presents many challenges due to the many variables involved, such as land use, animal husbandry, use of electricity, and fuels. One of the main challenges is to deal with this data heterogeneity. The data landscape presents obstacles such as managing source diversity, handling massive data volumes, adapting to various data formats, and ensuring real-time integration for decision-making. In this complex data landscape, ontologies emerge as a potential solution. An ontology is defined as a formal representation of a set of concepts within a domain and the relationships between those concepts. This study proposes an ontological model called CarbOnto for the syntactic and semantic integration of heterogeneous databases. Using an ontology, we intend to contribute to the standardization and interpretation of domain concepts and the addition of semantic information to generate complete GHG inventories. CarboOnto provides the means to generate farm inventories, identify imbalances, and search for solutions to neutralize gas emissions. We report a Case Study and argue that using this ontology can support balancing these gases.

INDEX TERMS Carbon balance, farm, GHG inventories, global warming, greenhouse gases, net zero, ontology.

I. INTRODUCTION

Global warming has attracted much attention in recent years. It is believed to be causing changes in weather patterns with consequent temperature rises. According to the Intergovernmental Panel on Climate Change (IPCC) of the United Nations (UN), these climatic changes are intrinsically associated with human activities, such as the increase in the emission of greenhouse gases (GHG) [1], [2], [3].

The Paris Agreement, signed in 2015 during the 21st Conference of the Parties (COP), established measures to reduce GHG emissions in response to the threats of climate change. The signatory governments committed to taking

mitigation actions to keep the average global temperature change below 2 °C compared to pre-industrial levels and to make efforts to limit the increase to 1.5 °C [3].

However, projections indicate that global warming of 1.5°C to 2°C will be surpassed during the 21st century unless drastic reductions in carbon dioxide (CO₂) and other greenhouse gas emissions occur in the coming decades. With more significant global warming, it is projected that each region will be more frequently subjected to simultaneous and multiple changes in climate impact agents, producing unpredictable effects for people and ecosystems [4].

Extreme weather events such as heat waves, droughts, winds, and heavy rains influence agricultural production. When intensified, the impacts on the agricultural sector

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could generate an increase in water needs for irrigation, an increase in the spread of diseases and pests of animals and crops, and a reduction in production, directly affecting societies and their life support activities, such as food production.

On the other hand, the agricultural sector is one of the main global emitters of GHG and therefore must take great steps towards mitigating global warming. Land use and land use change (LUC), caused by the conversion of native habitats to farmland, is considered the second largest source of greenhouse gas emissions on the planet, representing approximately 23% of the total [2], [5].

The agricultural sector is complex, with a significant diversity of production systems directly related to many environmental, social, and economic aspects [6]. Agricultural practices require a substantial amount of energy due to the use machine. Care is needed in land use and livestock farming to find solutions to implement that balance food production with sustainable development. Increasing production, increasing energy use efficiency, and reducing emissions related to agricultural production should be the desired results [7].

Net zero carbon emissions, or simply net zero emissions, is the state in which the net amount of GHG emissions released into the atmosphere is zero. The aim is to balance gas emissions and their removal so that the net balance is zero. We must align net-zero emissions with broader sustainable development goals, which entail an equitable transition to net-zero emissions, socio-ecological sustainability, and the pursuit of broad economic opportunities [2], [5].

In the search for net zero emissions, integrating databases from public sources with databases available on farms can generate knowledge. This knowledge can help build GHG balance inventories that can support decision-making in agriculture.

There are, however, challenges in calculating greenhouse gas emissions on rural properties, as there are many variables involved, and there needs to be more methods for calculating a complete inventory for an entire farm.

To tackle this, the Research Question addressed in this article is “*How can data integration support the GHG balance and knowledge generation of best practices towards net zero on farms?*”.

As a specific contribution, this research proposes an approach for the syntactic and semantic integration of heterogeneous databases related to sources of GHG emissions and stocks on farms and we present:

- An ontological model for data integration focusing on the standardization and interoperability of information between measurement systems.
- The discovery of new semantic relationships not naturally detected by inferences and SWRL (Semantic Web Rule Language) rules.
- Use of a semantic model (ontology) in a data preparation process encompassing data cleaning, normalization,

filling in missing data, and highlighting essential resources to be used by ML techniques.

This paper is organized into the following sections in addition to the Introduction. Section II addresses concepts related to carbon inventory and MRV (Measure, Report, and Verify) systems. Section III presents the related work. Section IV describes the adopted methodology. Section V details the CarbOnto ontology. Section VI presents a use case on a Brazilian farm. Section VII describes some research limitations, and Section VIII presents conclusions.

II. BACKGROUND

The Kyoto Protocol of the United Nations Framework Convention on Climate Change (UNFCCC), adopted in 1997, was the first international protocol to define flexibility mechanisms to help countries achieve their targets for reducing GHG emissions [8].

Among these mechanisms, the Clean Development Mechanism (CDM) stands out. The CDM allows countries to finance or develop GHG reduction projects outside their territory, negotiating in international markets. To achieve this, emissions reductions must be certified, bringing real, measurable, and long-term benefits [9]. Emission reductions aim to promote sustainability and its effects on climate change mitigation.

The Intergovernmental Panel on Climate Change [2], the United Nations body that deals with issues related to climate change, develops methodological reports that provide guidelines for preparing national GHG inventories. Within these guidelines, the concepts of “tiers” were established, representing the methodological complexity level adopted in the countries’ inventories. Usually, three tiers are provided. **Tier 1** is recommended for situations where country-specific emissions data are not available. IPCC suggests providing default data for estimates in these situations. **Tier 2** is recommended for situations where specific emission data or more refined empirical estimates are available for the country with some level of detail of the activities involved. **Tier 3** refers to using methodological procedures developed specifically by the country, including modeling and greater detailing of inventory measures [10].

The GHG Protocol [11] divides emissions into three scopes, classified according to the farm’s degree of responsibility or control over the sources of the emissions. Scope 1 includes direct emissions from sources on the farm or controlled by it. Sources are classified into mechanical, which consume fuel or electricity (tractors, trucks for example); non-mechanical, which emit GHG through biochemical processes (enteric fermentation of cattle and soil liming for example); and changes in land use and ground cover. Scope 2 includes indirect emissions from acquiring electrical and thermal energy purchased and consumed by the farm. Scope 3 includes all other indirect emissions not reported in Scope 2 that occur in sources that do not belong or are not controlled

by the farm, such as gases generated in transporting and storing the produced items.

To achieve the results, countries need to Measure, Report, and Verify (MRV) reductions. An MRV system consists of processes and procedures through which information is provided, evaluated, and verified to determine “if, how and how much” the parties involved effectively fulfilled their obligations [12]. Measurement is needed to identify emission trends and determine where to focus efforts. At the same time, reporting and verification are important to ensure transparency, good governance, accountability, and the credibility of results [13].

In this vein, Decision Support Systems (DSS) at the farm level drive decisions that aim to reduce emissions attributable to site-specific agricultural practices. Although not all impacts are perceived at a local scale, decisions made locally influence global impacts [14].

Detailed carbon inventories on farms allow you to quantify the carbon stock in different stores, such as soil and vegetation biomass. They also enable the quantification of emissions from activities, such as livestock farming and fuel and electricity use. Inventories are basic requirements for developing an MRV system [9].

Considering the requirements of MRV systems, we therefore believe that an ontology can contribute to data integration and interoperability.

An ontology is an explicit specification of a conceptualization [15]. A conceptualization is an abstract view of the world that we hope to represent for some purpose and it encompasses a representation, formal naming, and definition of the categories, properties, and relations between the concepts, data, and entities that substantiate one, many, or all domains of discourse. OWL and SWRL are commonly used to implement ontologies. OWL is an ontology language for the Semantic Web with formally defined meaning. OWL ontologies provide classes, properties, individuals, and data values and are stored as Semantic Web documents. The SWRL is a language for the Semantic Web that can be used to express rules logic combined with OWL.

The knowledge generated with the integrated data (in an ontology) makes it possible to identify stock and emission estimates of carbon on farms. This model is part of an architecture [16], from a holistic viewpoint, using an ecosystem for developing net zero applications.

III. RELATED WORK

Arulnathan et al. [14] present a systematic review of decision support tools focusing on farm sustainability. Their study reviewed 19 applications characterizing and identifying trends in the methodological choices made by developers. All of them included estimates of GHG emissions with limited scopes.

Among the tools evaluated in [14], Ofoot stands out. The Ofoot tool [17] is presented as a system for calculating estimates of greenhouse gas emissions on organic farms.

It consists of an application that generates inventories related to emissions from equipment, infrastructure, and consumable materials used on the farm. The use of the software is limited to the cultivation of organic foods in the Pacific Northwest region of North America due to the parameterization of the estimation models.

Some studies and projects [18], [19], [20], have developed methodologies for calculating GHG emissions, but they focus on something other than farm inventories. Harris et al. [18] integrate data to map annual forest-related GHG emissions and removals worldwide, using IPCC guidelines as a methodological framework. The Social Carbon Project [19] developed a methodology for carbon control in native vegetation areas located on private properties, specifically aimed at afforestation and reforestation projects. The MapBiomass Project [20], using regression techniques, maps the organic carbon stocks of the soil in Brazil, covering the period from 1985 to 2021.

Thumba et al. [21] present a review of studies on decision support systems for mitigating GHG emissions in livestock farming. The scope of these studies is limited to raising livestock on farms, not accounting for other emission sources.

Each of these studies [14], [17], [18], [19], [20], [21] has its context and relevance. However, our study focuses on farms and proposes the creation of more complete GHG inventories with all their emission and removal sources.

Considering the use of ontologies as a semantic model, we consider that an ontology can directly contribute to achieving this CHG inventory. Some reasons for developing an ontology include sharing a common understanding of information structure between people or software agents, reusing and analyzing domain knowledge [22], [23] and defining a common vocabulary for those who need to share information [15]. In our study, we go one step further in proposing an axiomatically rich ontology, which allows the discovery of new implicit relationships between data. It facilitates integration, as will be presented in the following sections.

The Soil Mission Support (SMS) document [22], developed to support the European Commission and the Mission Board of Horizon Europe, aims to be a reference for creating soil and land management ontologies. The document is intended to implement a common language for sharing information related to soil health and land management. The paper, however, does not display ontological models or go into implementation details.

Kim and Baumann [24] suggest using ontologies to create smart contracts in agricultural systems’ measurement, reporting, and verification (MRV). The goal is to support smart contracts on different blockchain platforms, providing standardization and sharing of concepts between MRV systems. The proposal only suggests the use without going into details of models or implementations.

Di et al. [25] and Zhu et al. [26] propose ontologies related to the life cycle of products with a focus on controlling

GHG emissions. Carbon emissions in the input artifacts are transferred to the final components. The proposals aim to record emissions data in the life cycle of products to build a labeling model according to these emissions.

Konys [27] proposes an ontology of sustainability assessment based on a comprehensive and multidimensional view, including social, environmental, economic, ecological, cultural, and institutional aspects. Hou et al. [28] and Zhang et al. [29] address the engineering domain, focusing on sustainable construction. Its ontology proposals help engineers reduce environmental impact and design more environmentally friendly structural components.

Daouadji et al. [30] present an ontology proposal focused on reducing indirect GHG emissions from information technology devices, improving energy efficiency, and concentrating on the semantics of energy-related resources and their properties. At the same time, Zhou et al. [31] propose an ontology for the ideal cutting tool configuration used in machining processes from the perspective of reducing carbon waste emissions and energy savings.

There are studies that use or reference ontologies [25], [26], [27], [28], [29], [30], [31] that address aspects related to sustainability focusing on different domains. However, we found no studies that propose ontologies related to climate issues encompassing farm GHG emissions. Moreover, no proposal provides an axiomatically rich ontology, which enables the discovery of new implicit relationships, as we do in our work.

Analyzing the literature, we can see that there are many challenges in calculating GHG emissions and stocks, mainly due to the different variables involved related to soil conditions, climate, vegetation, animal husbandry, use of fuels, and use of electrical and thermal energy. The quality, integrity, methodological complexity, and data sources each country uses vary [18]. Solutions prioritize methodological issues to meet their specific objectives [14].

Our study proposes a global, transparent, and independent solution using a methodology capable of generating complete GHG inventories for farms as long as the datasets are available. Furthermore, with the results obtained, we hope it will be possible to verify, at a macro level, the collective impact of climate policies related to agriculture.

IV. METHODOLOGY

Considering the Intergovernmental Panel on Climate Change [2] tiers, discussed in section II, our proposal is related to Tier 3, namely creating a methodology for use in any region or country with available farm datasets. In this study, we use datasets from Brazilian farms.

To account for gas emissions, we follow the GHG Protocol Agricultural Guidance, developed by the World Resources Institute (WRI) [11], together with government and partner institutions, to promote the use of internationally accepted best practices for developing agricultural GHG inventories. Our study encompasses Scopes 1 and 2, which farms directly

control. We discuss the control of anthropogenic emissions of CO₂, CH₄, and N₂O gases, as these gases are typically related to the activities of an agricultural chain. The metric used to equalize the emissions of these gases is carbon dioxide equivalent (CO₂e), based on the relative importance of each one in producing a certain amount of energy (per unit area), several years after an emission boost [11].

To calculate the GHG inventory of farms, we use an ontological model for syntactic and semantic integration, the CARBONTO ontology (section V). To develop the ontology, the following phases were executed [32], [33]: (i) Specification, (ii) Conceptualization, (iii) Formalization, and (iv) Evaluation. In the specification phase we identify the purpose, scope, implementation language, and intended End-Users and Competence Questions (CQ) (see section V). CARBONTO was developed to support syntactic and semantic integration of data (purpose) and the scope was defined considering the net zero domain (scope). The OWL 2.0 and SWRL languages were used as implementation languages. The intended users are farmers and researchers interested in the net zero domain.

The conceptualization phase focused on organizing and structuring the semantic meaning of data. This phase was based on the specification of terms and the relationships defined between these terms. The formalization phase was carried out using the Protegé tool and OWL and SWRL languages. We also defined a set of rules to support the semantic processing of the terms and discover new relationships between them. Finally, the evaluation phase consisted of carrying out verification and validation. The verification step consists of verifying the correctness and validating the ontology. The correctness of the ontology is done through a verification process using the Pellet plugin reasoner. The validation is a step to ensure that the ontology fulfills its purpose through answering Competency Questions (CQ). With the ontology formalized and validated, CARBONTO was used to process the syntactic and semantic integration of data. Brazilian specialists in the carbon market validated the CarbOnto ontology.

At the syntactic level, ontology classes and SWRL rules contribute to the preparation and persistence of data according to an integrative model, preserving the consistency and integrity of the database. At the semantic level, the use of ontologies is essential to make these applications interoperable. The diversity of views on the domain associated with the variety of possible representations of entities and their properties leads to difficulties in exchanging data in applications [34]. Ontology theory [35] supports the idea of a model that provides a formal and explicit representation of the meaning of vocabulary, which software applications can use as a reference for interoperability or development of new features. The ontology also allows us to find important relationships that are not naturally detected and infer specific situations by analyzing data through processing rules. Well-founded ontologies act as standards for the use of terminology and also follow strict rules for evolution and

reuse, leading to an ecosystem of potentially interoperable artifacts [34], [36].

In the following section, we present CarbOnto, the ontological model proposed to integrate the data.

V. CarbOnto

CarbOnto is an ontology for integrating heterogeneous bases related to GHG emitting and sequestering sources in rural properties. CarbOnto derives knowledge of relationships to generate a GHG balance on farms.

As presented in the previous section, an ontology model is based on Competency Questions (CQ) to accommodate the specific domain necessities. A CQ is a natural language sentence expressing a pattern for a question that people/computational applications expect an ontology to answer [37]. We elaborated on these CQ based on interviews with researchers and farmers and documents related to gas emissions. We used ontological concepts, including their classes, relations, and inference rules, to answer the CQ. The Competency Questions describe what was expected from the ontology to answer. The main ontology competency questions related to CarbOnto specification are: (i) What are the sources of GHG emission and sequestration from a farm? (ii) How many cultivation areas is the farm divided into? (iii) What are the GHG emission and sequestration values for each area? (iv) How many ruminant animals are there on the farm? (v) What are the breeds, sex, age, and number of animals on the farm? (vi) How much CH₄ is emitted by the animals? (vii) What are the existing mechanical emission sources? (viii) How much CO₂ is emitted by mechanical sources? (ix) What are the existing non-mechanical sources of emissions? (x) How much CO₂ or N₂O is emitted by non-mechanical sources? (xi) What is the total amount of GHG emissions of the farm? (xii) What total amount of GHG does the farm sequester? (xiii) What is the farm's GHG balance? To answer the CQs, we have to process specific rules over ontological data. Therefore, we need an axiomatically rich model to answer these questions.

The data integration and the processing of specific semantic rules should enable answers to these competence questions. The expected result is the GHG balance or inventory.

Farms can have diverse reasons for developing inventories. These generally involve (i) identifying opportunities to reduce GHG emissions or sequestration, setting baselines and reduction targets, and tracking performance; (ii) identifying opportunities to reduce costs and increase productivity; (iii) managing reputational risks and opportunities associated with agricultural GHG fluxes. These reasons can help reduce costs and increase farm productivity [11]. GHG emissions reduction measures may also offer co-benefits such as reduced erosion and land degradation, improved water quality and retention, control of air pollutants, and increased soil fertility.

Only some studies have analyzed the relative contribution of different sources to inventories of different agricultural systems using a consistent set of methods.

It is a complex job [11]. Measurement can be hampered by the inability to account for the quantity and impact of emissions transparently and uniformly.

The role of ontologies, in this context, would be to contribute to the necessary standardization, sharing, and interoperability of data [24]. An ontological model can contribute to selecting, integrating, and filling in missing data. This can be done by processing new relationships between data and gaining knowledge from discovering these new relationships. Furthermore, it can provide data quality, a critical factor for the accuracy of artificial intelligence (AI) techniques [38].

International guidelines, protocols, and methodologies must guide a reliable measurement process. This study is based on IPCC guidelines [2] and WRI protocols [11]. The integrated datasets should be provided by governmental and non-governmental organizations and private institutions that use public collection, processing, and storage protocols, preferably published in articles in recognized scientific journals. Figure 1 presents the CarbOnto ontology, comprising the main classes and their relationships. Table 1 describes these classes. The ontology is available at <https://github.com/lfs19>.

These classes are used to support data integration. We use the bottom-up strategy for data integration [39]. Information on GHG emissions should be collected at the lowest possible organizational level of accounting and quantification [40]. The lowest level considered in this study is represented by areas, which are portions of crops within rural properties. Once the GHG emission estimates have been made in each area, they are consolidated with the other data generated by other sources, generating estimates for the entire farm.

The bottom-up integration process is made up of two steps: translation and schema generation. In the first step, the dataset schemas are translated into a common intermediate canonical representation, the CarbOnto ontological model, and in the second step, the intermediate schema is used to generate a global conceptual schema [39].

For the development of CarbOnto, we used Protégé 5.5.0 software,¹ for the ontology specification and the OWL-Ready2² framework for the implementation. We also used Pellet³ as an ontology reasoner and SWRL⁴ to create inference rules. CarbonOnto is flexible and scalable, i.e., it can easily extend the model for specific niches and can accommodate a large number of classes because the ontology instantiation is executed using the Persistent World (PW) concept,⁵ which enables the processing of large volumes of data in ontologies. The ontology access is performed from a component that uses the OwlReady2 framework in Python. This approach enables OwlReady2's optimized quadstore

¹<https://protege.stanford.edu>

²<https://owlready2.readthedocs.io/en>

³<https://github.com/stardog-union/pellet>

⁴<https://www.w3.org/Submission/SWRL>

⁵<https://owlready2.readthedocs.io/en/latest/world.html>

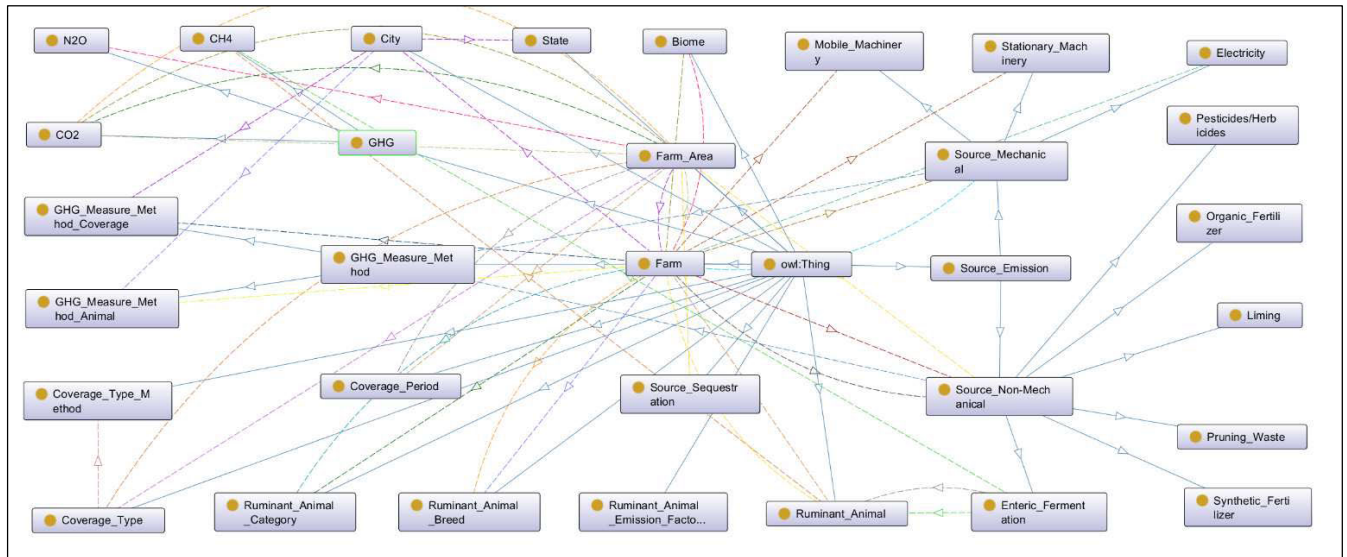


FIGURE 1. CarbOnto classes and relationships.

to be stored in an SQLite³⁶ database. The ontology has an optimized processing performance with the PW approach to load and execute inferences. Thus, this approach divides the ontology into a model (TBox – only model description) and a complete ontology (Tbox + Abox, description of model and instances). From Tbox, a world is created to receive a limited number of instances and limit the reasoner’s processing time (the architecture currently uses the Pellet reasoner). After inference processing, this processed new world is loaded into the complete ontology for queries. PW allows the persistence of all the worlds of ontologies created and loaded.

VI. CASE STUDY USING BRAZILIAN FARMS

Brazil is a leading agricultural producer. In 2020 Brazilian cattle accounted for about 14% of the world's herd. Brazilian grain production was the fourth largest in the world, representing approximately 8%, according to Food and Agriculture Organization Statistics [41]. In contrast, Brazil presents high heterogeneity in land use patterns, agricultural management practices, and carbon stocks. Land use and land use change (LUC) accounted for the largest share of GHG emissions in 2021, corresponding to 46% of the gross total [42]. Due to the importance of agricultural activity in the Brazilian economy, this sector is a fundamental part of the strategy to mitigate the emission of greenhouse gases.

There is a diversity of data, mainly on public databases, which, once integrated, make it possible to calculate the GHG balance. This can generate knowledge in the search for best practices related to crops and animal husbandry on farms so as to evaluate the agricultural practices as ecologically correct.

This case study aims to evaluate the following Research Question (RQ): “*How does data integration support the GHG balance and knowledge generation of best practices towards net zero on farms?*”.

A. EMISSION ESTIMATES AND CARBON STOCK BY LAND USE

In this case study, we use the following datasets: Land Tenure Map [43], Land Use Map [44], Brazilian Land Use Change Method (BRLUC) [45], CH4 Emission Factors for Cattle [46]. The Land Tenure Map [43] integrates several Brazilian public governmental databases with georeferenced polygons of rural properties. The MapBiomass Project [44] maps the land cover of the country. This is an initiative of some teaching and research institutions to generate maps of land use and land cover from automatic classification processes applied to satellite images.

Brazilian Land Use Change [45] is a method for calculating estimates of direct changes in land use and CO₂ emissions associated with Brazilian agricultural products, developed by researchers from the Brazilian Agricultural Research Corporation (EMBRAPA). The method is compatible with the main international Life Cycle Assessment and Carbon Footprint protocols. The enteric fermentation methane emissions factors for cattle [46] is a publication of the General Coordination of Global Changes of the Brazilian Ministry of Science and Technology. Figure 2 shows an overview diagram of the data integration model. A wrapper component was defined for exporting data from the source schema for each integrated database.

In Google Earth Engine,⁷ the Land Use Map, which contains data on the land use and cover of the country in 2021,

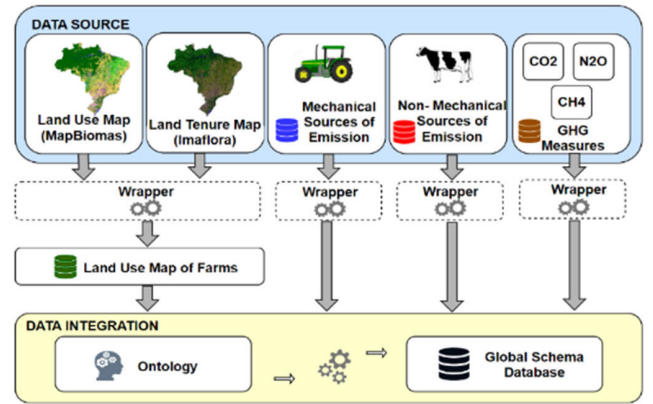
⁶<https://www.sqlite.org>

⁷<https://earthengine.google.com>

TABLE 1. Main classes of CarbOnto.

Farm: Represents rural properties.
Biome: Represents the biome in which the farm is located.
City: Represents the city where the farm is located.
State: Represents the city location state.
Size: Refers to the size of the rural property.
Greenhouse Gases (GHG): The amount of greenhouse gases emitted or sequestered by the farm's mechanical and non-mechanical sources, typically related to the activities of an agricultural chain: CO ₂ , CH ₄ , N ₂ O.
GHG Measure Method: These are the methods used to account for GHG emissions.
Areas: Cultivation areas (cover and land use) within each rural property. The area was defined, in this methodology, as the lowest organizational level of accounting for emission and stock of greenhouse gases on a farm. The sum of the areas results in the size of the farm.
Coverage Period: Number of months of the year used for a given crop. The reference period for calculating the GHG balance is one year. Cropland can be used for crop rotation, with more than one type of land use during the year.
Coverage Type: Vegetation cover represents the land use in each area within a farm. Each type of coverage (forest, pasture, soy, coffee, etc.) or even its absence (in the case of degraded soils) can present the soil's emission values or carbon stock.
Coverage Type Method: Method used to identify land cover areas. It can be through satellite images or site surveys using drones.
Sequestration Source: Represents the sources of GHG sequestration which, in the case of farms, represents the crops or plantations that sequester CO ₂ .
Emission Source: Represents the sources of GHG emissions on a farm, divided into (1) mechanical and (2) non-mechanical sources.
(1) Mechanical Source: Relating to equipment and machinery operated by the farm. CO ₂ emissions from such direct or indirect sources must be classified into categories (subclasses)
(i) Stationary Machinery: Pressing, drying, processing equipment, etc. (ii) Mobile Machinery: Sowing machine, combine harvester, tractors, etc. (iii) Electricity: Indirect emissions from the purchase and use of electricity on the farm.
(2) Non-Mechanical Source: Relative to the other emission sources and must be classified in (i) Enteric Fermentation: CH ₄ emissions resulting from the enteric fermentation process of ruminant animals. (ii) Liming: N ₂ O emissions from using limestone in the soil. (iii) Synthetic Fertilizer: N ₂ O emissions resulting from the use of synthetic fertilizers. (iv) Organic Fertilizer: CO ₂ emissions from using organic fertilizers (animal waste and crop residues deposited on the soil). (v) Pesticides/Herbicides: N ₂ O emissions resulting from pesticides and herbicides in plantations. (vi) Pruning Waste: CO ₂ emissions resulting from pruning residues from plantations.
Ruminant Animal: Refers to ruminant animals that emit CH ₄ . Animals do not need to be computed by area, they can be counted at the farm level, considering that they can be raised in confined areas or areas integrated with other crops, for example, the areas of integration between livestock, crops, and forest.
Animal Category: Categories of animals raised, such as cattle, goats and sheep.
Animal Breed: Breed of raised animals related to CH ₄ emission.
Animal Quantity: number of animals raised.
Animal Emission Factor: CH ₄ emission factor of animals related to factors such as age, sex, and milk production.

and the Land Tenure Map, which includes the georeferenced polygons of Brazilian rural properties, were superimposed. After superimposing these maps, scripts extract the land use areas for each rural property.


FIGURE 2. Data integration diagram.

The following images show farms in the state of Minas Gerais, Brazil. We chose this state because it has various land cover and land use patterns. Figure 3 illustrates, in a blue line, the limits of Minas Gerais state inside Brazil.

Figure 4 shows the farm polygons with their respective vegetation cover from automatic classification processes applied to satellite images. Based on a stratified random sample of 75,000 pixel locations, the average overall accuracy of this classification was 89%, ranging from 73 to 95% in the Brazilian biomes [47]. Images may contain clouds, smoke, and other artifacts that make them blurry. To produce a clean image, cloud-free pixels are preferably selected from the available images; otherwise, the images need to be treated with dehazing techniques [48], [49], [50].

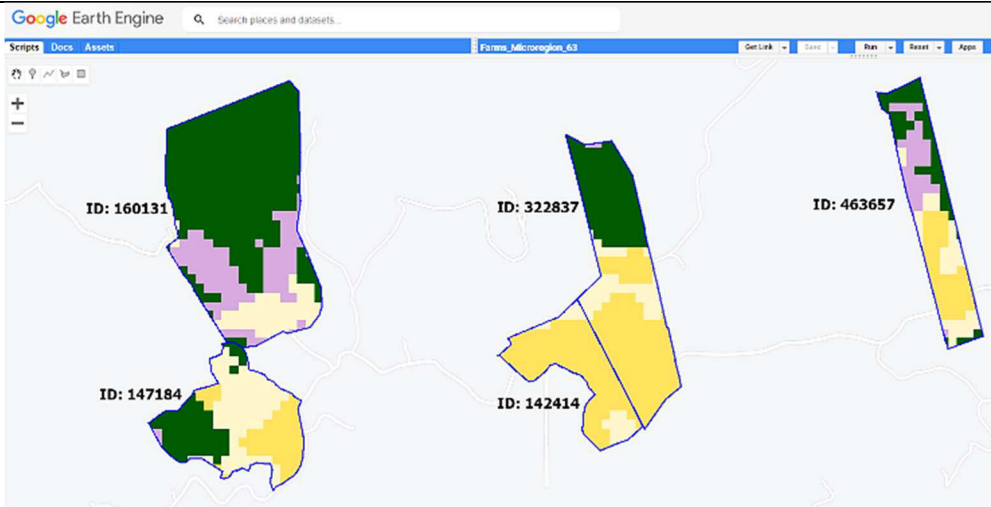
From the farm identification, we divided emissions and CO₂ stock calculation areas according to the use and land cover.

Estimated data of CO₂ emissions associated with 64 types of crops, planted pastures, and forestry in Brazilian municipalities were obtained using the BRLUC Method⁸ and integrated by the ontology.

The integration of these data allows us to know the stock and CO₂ emission estimates for the analyzed farms. For example, Table 2 shows the land cover map of the farms identified with the codes: 160131, 463657, 147184, 142414, and 322837 (we are using the same code as the Land Tenure Map), located in the municipality of Pedra Dourada, state of Minas Gerais, Brazil.

To help answer our RQ, regarding carbon emission by land use, we processed specific queries over the CarbonOnto ontology. CarbonOnto's SWRL rules for calculating land cover use 62 concepts from the Mapbiomas Project [44] and helped to answer the following competency questions: "(ii) How many cultivation areas is the farm divided into?" and "(iii) What are the GHG emission and sequestration values of each area?". Table 3 presents the SWRL rules for calculating CO₂ stock and emissions for the farm areas.

⁸<https://brluc.cnpma.embrapa.br>

TABLE 2. Emission estimates and carbon stock from the areas of the farms.


Coverage codes	Color	Legend Classes	Area (ha)	Estimates of carbon emission by hectare in tCO ₂ .ha ⁻¹ .yr ⁻¹ *	Estimates of carbon stock by hectare in tCO ₂ .ha ⁻¹ .yr ⁻¹	Estimates of carbon emission by area in tCO ₂ .ha ⁻¹ .yr ⁻¹	Estimates of carbon stock by area in tCO ₂ .ha ⁻¹ .yr ⁻¹	Positive carbon balance (stock - emission)
Farm ID: 160131								
3	Green	Forest Formation	21,72	0	150,36	0,00	3265,82	3265,82
9	Yellow	Forest Plantation	0,06	2,36	65,33	0,14	3,92	3,78
21	Orange	Mosaic of Uses	3,78	0,64	45,72	2,42	172,82	170,40
46	Pink	Coffee	7,02	0	69,87	0,00	490,49	490,49
Total						2,56	3933,05	3930,49
Farm ID: 147184								
3	Green	Forest Formation	3,95	0	150,36	0,00	593,92	593,92
15	Yellow	Pasture	3,81	0,64	45,72	2,44	174,19	171,75
21	Orange	Mosaic of Uses	5,64	0,64	45,72	3,61	257,86	254,25
46	Pink	Coffee	0,12	0	69,87	0,00	8,38	8,38
Total						6,05	1034,36	1028,31
Farm ID: 322837								
3	Green	Forest Formation	6,06	0	150,36	0,00	911,18	911,18
15	Yellow	Pasture	9,03	0,64	45,72	5,78	412,85	407,07
21	Orange	Mosaic of Uses	2,58	0,64	45,72	1,65	117,96	116,31
46	Pink	Coffee	0,08	0	69,87	0,00	5,59	5,59
Total						7,43	1447,58	1440,15
Farm ID: 142414								
15	Yellow	Pasture	7,49	0,64	45,72	4,79	342,44	337,65
21	Orange	Mosaic of Uses	1,32	0,64	45,72	0,84	60,35	59,51
Total						5,64	402,79	397,15
Farm ID: 463657								
3	Green	Forest Formation	2,94	0	150,36	0,00	442,06	442,06
15	Yellow	Pasture	3,34	0,64	45,72	2,14	152,70	150,57
21	Orange	Mosaic of Uses	2,49	0,64	45,72	1,59	113,84	112,25
46	Pink	Coffee	2,58	0	69,87	0,00	180,26	180,26
Total						3,73	888,87	885,14

*tCO₂.ha⁻¹.yr⁻¹ = ton of carbon per hectare per year

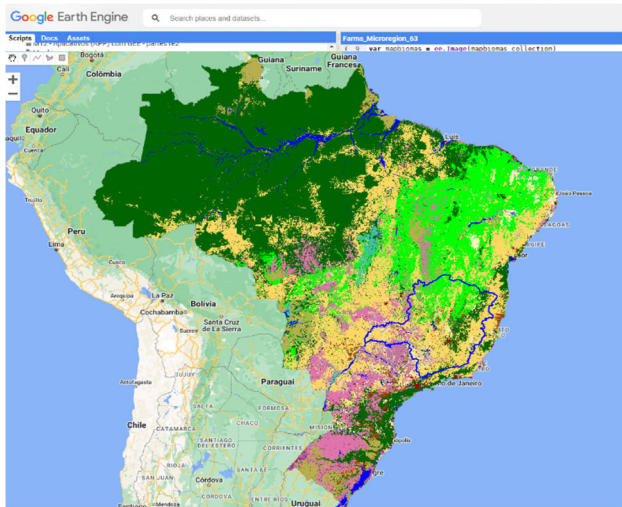


FIGURE 3. Land cover and land use map of the Brazilian territory with outline in blue line of the state of Minas Gerais.

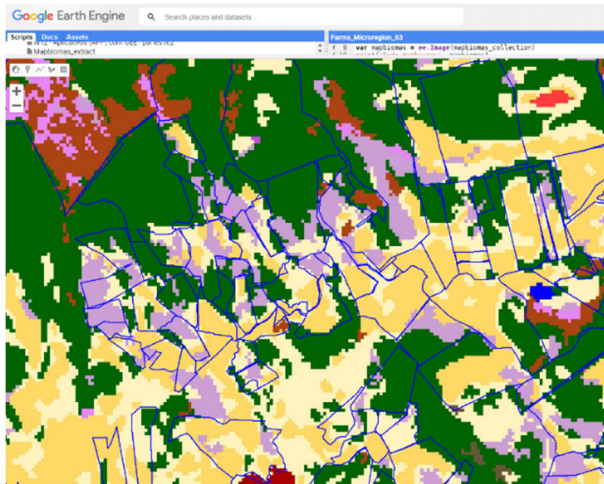


FIGURE 4. Outline of the polygons of farms in the state of Minas Gerais showing land use and land cover.

With the estimates for each area, we consolidate the total value of the farm and calculate the carbon balance. The carbon balance for the five farms analyzed was positive; the farms have the following carbon stocks in $\text{tCO}_2\text{ha}^{-1}\text{yr}^{-1}$ (tons of carbon per hectare and year): Farm ID:160131 (3930,49), Farm ID: 147184 (1028,31), Farm ID: 322837 (1440,15), Farm ID: 142414 (397,15) and Farm ID: 463657 (885,14).

Calculated estimates may differ from field-measured values. However, it is important to highlight that the GHG estimates for most Brazilian farms, such as those we used in the study, are unknown. Current carbon measurement methodologies and techniques are inaccessible due to cost, especially for small rural producers. In addition, it is important to highlight that the data used in this study were carefully selected from studies and methodologies developed

TABLE 3. SWRL rules used for calculating carbon balance.

Name: Emission Value CO2 per Area
$\text{swrlb:multiply}(\text{?total}, \text{?hsfa}, \text{?heffa}) \wedge \text{hasSizeFarmArea}(\text{?fa}, \text{?hsfa}) \wedge \text{hasEmissionFactorCO2FarmArea}(\text{?fa}, \text{?heffa}) \wedge \text{Farm_Area}(\text{?fa}) \rightarrow \text{hasEmissionValueCO2FarmArea}(\text{?fa}, \text{?total})$
Name: Sequestration Value CO2 per Area
$\text{swrlb:multiply}(\text{?total}, \text{?hsfa}, \text{?hsffa}) \wedge \text{hasSizeFarmArea}(\text{?fa}, \text{?hsfa}) \wedge \text{hasSequestrationFactorCO2FarmArea}(\text{?fa}, \text{?hsffa}) \wedge \text{Farm_Area}(\text{?fa}) \rightarrow \text{hasSequestrationValueCO2FarmArea}(\text{?fa}, \text{?total})$
Name: Balance CO2 per Area
$\text{swrlb:subtract}(\text{?total}, \text{?hsyfa}, \text{?hevfa}) \wedge \text{hasSequestrationValueCO2FarmArea}(\text{?fa}, \text{?hsyfa}) \wedge \text{hasEmissionValueCO2FarmArea}(\text{?fa}, \text{?hevfa}) \wedge \text{Farm_Area}(\text{?fa}) \rightarrow \text{hasCO2BalanceFarmArea}(\text{?fa}, \text{?total})$

by research institutions and groups of researchers with significant expertise in their respective areas, supported by scientific publications [20], [44], [45].

B. ESTIMATES OF CH₄ EMISSION BY THE CATTLE HERD

Data on CH₄ emissions from bovine enteric fermentation was obtained from the publication of the General Coordination of Global Changes of the Brazilian Ministry of Science and Technology⁹ [46]. The number of cattle per farm, however, needs to be included. Brazil has no national control over the number of livestock distributed on rural properties. The numbers are dynamic and constantly changing. We therefore intend to integrate data from smart farms and other farms that control livestock through microchips as soon as this data is available.

To help answer the RQ, we used the SWRL rules to compute the CH₄ estimates. Our specific SWRL rules helped to answer the following competency questions: “(iv) How many ruminant animals are there on the farm?”, “(v) What are the breeds, sex, age, and number of animals on the farm?” and “(vi) How much CH₄ is emitted by animals?”.

The calculation of emissions due to enteric fermentation [51] is carried out using the formula:

$$\text{EmissionCH}_4 = \text{NAXEFCH}_4, \text{ where:}$$

- EmissionCH₄= CH₄ emission associated with enteric fermentation (kg de CH₄/year)
- NA = number of animals
- EFCH₄ = CH₄ emission factor for enteric fermentation

Table 4 displays the SWRL rules used to calculate animal enteric fermentation.

⁹<https://bit.ly/2011mct>

TABLE 4. SWRL rules used for calculating animal enteric fermentation.

Name: Emission CH4 Cattle Male
$Farm(?f) \wedge hasCattleMaleQuantity(?f, ?hcmq) \wedge swrlb:multiply(?total, ?hcmq, ?efCHCattleMale) \rightarrow hasEmissionValueCH4CattleMale(?f, ?total)$
Name: Emission CH4 Cattle Female
$Farm(?f) \wedge hasCattleFemaleQuantity(?f, ?hcfq) \wedge swrlb:multiply(?total, ?hcfq, ?efCHCattleFemale) \rightarrow hasEmissionValueCH4CattleFemale(?f, ?total)$
Name: Emission CH4 Cattle Female Milk
$Farm(?f) \wedge hasCattleFemaleMilkQuantity(?f, ?hcfmq) \wedge swrlb:multiply(?total, ?hcfmq, ?efCHCattleFemaleMilkQuantity) \rightarrow hasEmissionValueCH4CattleFemaleMilk(?f, ?total)$
Name: Emission CH4 Cattle Young
$Farm(?f) \wedge hasCattleYoungQuantity(?f, ?hcyq) \wedge swrlb:multiply(?total, ?hcyq, ?efCHCattleCattleYoung) \rightarrow hasEmissionValueCH4CattleYoung(?f, ?total)$
Name: Animal Total Quantity
$Farm(?f) \wedge hasCattleFemaleMilkQuantity(?f, ?hcfmq) \wedge hasCattleMaleQuantity(?f, ?hcmq) \wedge hasCattleFemaleQuantity(?f, ?hcfq) \wedge hasCattleYoungQuantity(?f, ?hcyq) \wedge swrlb:add(?total, ?hcyq, ?hcmq) \wedge swrlb:add(?total2, ?total, ?hcfq) \wedge swrlb:add(?total3, ?total2, ?hcfmq) \rightarrow hasTotalAnimalQuantity(?f, ?total3)$
Name: Emission CH4 Total
$Farm(?f) \wedge hasEmissionValueCH4CattleYoung(?f, ?hevcy) \wedge hasEmissionValueCH4CattleMale(?f, ?hevcem) \wedge hasEmissionValueCH4CattleFemale(?f, ?hevcf) \wedge hasEmissionValueCH4CattleFemaleMilk(?f, ?hevcfm) \wedge swrlb:add(?total, ?hevcfm, ?hevcf) \wedge swrlb:add(?total2, ?total, ?hevcem) \wedge swrlb:add(?total3, ?total2, ?hevcy) \rightarrow hasEmissionTotalValueCH4(?f, ?total3)$

C. ELECTRICITY EMISSION ESTIMATES

To estimate electricity emissions, obtaining the average CO₂ emission factors is necessary. These factors estimate the amount of CO₂ associated with a given amount of electricity generated. The factor results from the average calculation of generation emissions, considering all generating plants. If all consumers calculated their emissions by multiplying the energy consumed by this factor, the sum would correspond to the emissions of the entire national system. Thus, it should be used when the objective is to quantify the emissions of electricity being consumed in each period.

The average CO₂ emission factors are calculated monthly and made available by the Brazilian Ministry of Science, Technology, and Innovation,¹⁰ under the guidelines of the UNFCCC.¹¹

¹⁰<https://bit.ly/3Xc15qw>

¹¹<https://bit.ly/PAMethodologies>

In this case, to help answer the RQ, the SWRL rule for calculating the farm's emissions due to the consumption of electricity [51] is displayed in Table 5.

TABLE 5. SWRL rule for calculating the farm's emissions from the consumption of electricity.

Name: Emission electricity
$Farm(?f) \wedge hasElectricityConsumption(?f, ?hec) \wedge swrlb:multiply(?total, ?hec, ?nef) \rightarrow hasEmissionElectricity(?f, ?total)$
Where, - hasEmissionElectricity is the emission of CO ₂ in ton. - hasElectricityConsumption is the energy consumption in MWh; - nef is National Emission Factor.

D. EMISSION ESTIMATES FROM OTHER SOURCES

Emissions calculations for some sources depend on the data made available by farms. Some mechanical sources, such as mobile and stationary machines, rely on fuel. Emissions for fertilizers, limestone, pesticides, and herbicides also depend on the amounts used by the farm.

Driven by the concept of smart farms, many of these calculations can be simplified using sensors through automated measurements. We cannot access these data, considering that the data from the farms used in this case study are not equipped with sensors. However, our recent studies [52] are considering smart farms, such as the compost barn system from EMBRAPA (Brazilian Agricultural Research Corporation). Although not yet conclusive, our findings are driving improvements in the results.

VII. CASE STUDY CONSIDERATIONS

The case study provided the following findings:

- In the absence of data, metadata related to land cover can be extracted from geospatial datasets to generate GHG estimates for land use.
- The inferences performed by the ontological model contributed to adding semantic information and discovering new relationships between data. These new relationships assist in inventory generation.
- Considering the aspects related to the standardization of terms in the agricultural GHG domain and the use of SWRL rules to calculate estimates, the CarbOnto ontology is suitable for use as a component of an MRV system or in systems to support farmer decision-making.

We did not find studies on ontologies built to address climate issues, emphasizing GHG inventories in agriculture. Thus, CarbOnto can fill this gap or integrate other existing or emerging ontologies in this domain.

CarbOnto's initial proposal is for use in an MRV system, which is composed of the integration of specific systems and the collaboration of different actors, including domain specialists and public and private organizations.

VIII. RESEARCH CHALLENGES AND LIMITATIONS

The main challenges for generating GHG inventories on farms are:

- (i) Lack of data on farms and in open databases related to electricity and fuel consumption, use of fertilizers, pesticides, and herbicides. Lack of detailed data on ruminant livestock, such as breed, gender, and age. Despite specific data lacking in the case study, the ontological model was prepared to process them and perform GHG estimate calculations when available and integrated.
- (ii) Land use and land cover data for the Brazilian territory are unavailable in open databases. However, we could extract them from geospatial datasets [44]. With these data and data from the BRLUC Method [45], we were able to generate the GHG balance for land use.
- (iii) Our literature review detected a lack of standardization in the terms used in the agricultural GHG domain and the emissions calculations. Using ontology as an integrative model allows to reduce or eliminate conceptual and terminological misinterpretation, unifying different points of view and serving as a basis for communication between people and interoperability between systems. A well-defined semantic structure can provide better system specifications and component reuse. Our ontological model proposes a methodology to standardize terms and calculations based on IPCC guidelines [2] and WRI protocols [11].

The generation of complete GHG inventories on farms depends on the availability of all the datasets specified in CarbOnto. For smart farms, the limitations decrease as IoT sensors can obtain this data. Farm owners can also include this data collaboratively.

IX. CONCLUSION AND FUTURE WORK

This paper presented a solution to investigate the following research question: “How can data integration support the GHG balance and knowledge generation of best practices towards net zero on farms?”. To tackle this RQ we propose an ontology, called CarbOnto, as a solution to integrate data from heterogeneous GHG emission and sequestration sources. We discussed a case study with integrated public datasets from institutions in Brazil and showed the land cover and use balance of a Brazilian farm. The results have helped to answer our RQ partially, considering that we only used data related to the use and cover of farm soils.

CarbOnto was specified with the aim of standardization, sharing, interoperability, and derivation of knowledge to generate inventories of greenhouse gases on farms. Additionally, our proposal aims to contribute to the review and readjustment of agricultural practices to more ecologically correct ones, considering that this sector is one of the main global emitters of GHG.

As future work, we intend to provide provenance data to help in data reliability and use machine learning techniques to help in intelligent processing. Provenance is intended to

capture the data history from its origin, enabling traceability and considering the importance of reliability and compliance requirements in the generated inventories. Machine learning techniques will be used to identify and extract patterns with a positive performance in the GHG balance, indicating best practices in land use and management, including the use of fertilizers, herbicides, and pesticides. It can also help to reduce emissions from mechanical sources. This data can serve farms that want to improve farming practices.

Nature-based solutions such as sustainable land management can significantly contribute to limiting global warming and collaborating with human societies to maintain livelihoods and food security. For farms, generating and certifying carbon credits could be an economic opportunity. CarbOnto collaborates, in this context, to develop sustainable agriculture through integrating agricultural data towards net zero.

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